

The Hand That Measures

*Greek δῶρον 2 and the Body-Part-as-Measure Cluster on PIE *de-r-*

Companion paper to *The Hand That Gives* (Paper 1 of the *der-* project)

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ABSTRACT

Frisk's *Griechisches Etymologisches Wörterbuch* and Beekes's *Etymological Dictionary of Greek* both treat Greek δῶρον as two entries: δῶρον 1 “gift,” with its established derivation from PIE **deh₃-ro-*, and δῶρον 2 “palm-of-the-hand, palm-as-measure,” marked by Frisk as origin unknown and by Beekes (2010: 363) as <?> (origin uncertain). The Hesychian gloss δῶρον · παλαιστή, the metrological compounds in *Iliad* Δ 109 and Hesiod *Works and Days* 426, the anatomical δῶρον of Pollux 2.144, and Vitruvius's *De architectura* 2.3.3 all attest the second word. What they do not supply is its etymology.

This paper supplies that etymology and locates δῶρον 2 within a body-part-as-measure cluster on the framework's PIE **de-r-* “arm, fore-arm, open palm” (the body-part heteroclite arm-noun of the companion paper *The Hand That Gives*). The derivation uses Greek-internal morphology: Szemerényi's Law on the consonant-stem r-stem nominative singular **do-r-s* yields the lengthened nominative **dōr* (parallel to εὐπάτωρ from **eu-patōr-s* and to Homeric δώτωρ “giver” from **deh₃-tōr-s*); subsequent thematization **dōr* → **dōr-o-n* surfaces as δῶρον 2. The same derivational scaffold generates the open-palm cluster: Hesychian δάρις “span” (i-stem on the bare e-grade), Old Irish *derna* “palm” (feminine singulative **nā-* on the e-grade), Continental Celtic DERNV[(Stifter 2022), and — defended in a standalone companion paper — Albanian *dorë* (e-grade Szemerényi plus **-eh₂-*). The relation between δῶρον 2 and δάρις is sibling-derivational, not phonological alternation: Schrijver's (1995) “obscure” verdict dissolves once the morphological independence of the two forms is recognized.

A parallel arm-as-measure cluster on the s-stem sister heteroclite **dóws-* generates Sanskrit *dos* = 18 inches (*Mātaṅgalīlā* 1.14) and *dor-jyā* “arm-string” = sine (*Sūryasiddhānta* 3.10), placed inside the near-universal Indo-European cubit family (Latin *cubitum*, Greek πῆχυς, Hebrew *ammah* “mother of the arm,” Old English *ell*). The body-part heteroclite class generates measurement vocabulary in both members: the r-stem yields palm/span-class measures (~3–9 inches), the s-stem yields cubit-class measures (~18 inches). The metrology is a productive feature of the class, not a one-off feature of either heteroclite. The

framework does not contest δῶρον 1 or its standard derivation. Vitruvius’s etymology (“the Greeks call the palm δῶρον because giving is called δῶρον”) is folk etymology of the standard kind, where speakers confronted with two homophones construct a semantic bridge to motivate the perceived connection. Probability conditional on Paper 1’s framework: 50–60%. Project-level marginal probability: 2–5%.

1. Introduction: The Two δῶρα

Greek δῶρον appears in the standard Greek etymological dictionaries as two separate entries. Frisk’s *Griechisches Etymologisches Wörterbuch* (1960–72) places δῶρον “Gabe, Geschenk” on one entry and δῶρον “Handfläche, Palme” on a separate entry marked “Origin: XX [etym. unknown],” while still noting the comparison to Albanian *dorë* via La Piana. Beekes (2010: 363) preserves the same two-entry structure: δῶρον 1 “gift” with its standard derivation from PIE **deh₃-ro-*, and δῶρον 2 “palm-breadth, palm-of-the-hand,” marked <?> (origin uncertain). Beekes’s ETYM section for δῶρον 2 surveys rejected comparanda without supplying a positive reconstruction. The two-entry treatment is not a framework innovation; it is the position the standard Greek etymological dictionaries already hold.

The lexicographic record for δῶρον 2 is rich and ancient. The Hesychian lexicon glosses δῶρον · παλαιστή — “palm-breadth,” the width of the four fingers held together. The metrological tradition preserves the unit in compound numerals: *Iliad* Δ 109 has ἑκκαίδεκάδωρος “sixteen palm-widths long” (of Pandaros’s bow horns), Hesiod *Works and Days* 426 has δεκάδωρος “ten palm-widths” (of an axle), Pollux *Onomasticon* 2.144 lists δῶρον as an anatomical term for the front of the palm, and inscriptions from Miletus preserve δῶρον as a standard architectural unit. The unit is well-attested across literary, lexicographic, anatomical, and epigraphic registers from the 8th century BCE forward.

Vitruvius, writing *De architectura* in the first century BCE, explains the unit’s name (2.3.3): the Greeks call the palm δῶρον because giving — *datio* — is called δῶρον in Greek. This is the only ancient explanation of the homophony to come down to us, and it is folk etymology: the kind of story an ancient writer would supply when confronted with two same-sounding words. Vitruvius’s story is semantically plausible — the HAND → GIVE grammaticalization documented by Heine and Kuteva (2002) is real — but as derivational evidence it is exactly what folk etymology always is: an attempt to make sense of synchronic homophony by inventing a diachronic link. English *ear* “organ of hearing” and *ear* “spike of grain” are two synchronically homophonous words with entirely different etymological histories; folk etymology links them. The same analytical posture applies to the two δῶρα.

This paper takes the two-entry treatment of δῶρον at face value and supplies the etymology Frisk left open. δῶρον 1 descends from the PIE give-verb **deh₃-* via the standard nominalization **deh₃-ro-* — an analysis the framework adopts without modification. δῶρον 2 descends from the body-part heteroclite arm-noun **de-r-* of *The Hand That Gives* (§5 below). The same root also produces Hesychian δάρις “span”, Old Irish *derna* “palm,” Continental Celtic DERNV[(Stifter 2022), and — developed in a standalone companion paper — Albanian *dorë*.

Three late-twentieth-century departures dismantled the Pokorny IEW 203 grouping that had united these forms. Schrijver (1995: 58) labeled the δωρ-/δαρ- relation “obscure.” Beekes (2010) excluded the Celtic forms on the ground that Celtic **u* “excludes any relationship with δῶρον.” Matasóvič (2009) classified Celtic **durno-* as a probable substrate borrowing. This paper addresses all three. The Schrijver “obscurity” resolves once the morphological independence of δῶρον 2 and δάρις is recognized (§3.5). The Beekes and Matasóvič objections target the fist-cluster (**durno-*, *dorn/dwrn*, Breton *dorn*, Latvian *dūre*), which the framework reassigns to PIE **deru-/*drū-* “firm, hard, oak, tree” (Paper 1 §7), removing the Celtic **u* problem from the present paper entirely (§8).

The comparative cluster around δῶρον 2 was assembled in stages. The Celtic-Greek palm comparison — Irish *derna* compared with Greek δῶρον in its palm-breadth sense — was already in the nineteenth-century work of Fick and Prellwitz. Pedersen (1909, vol. I §32) listed the gift-cluster. Marstrander (1919: 128) added the decisive Anatolian piece. Pokorny canonized the synthesis in IEW 203 in 1959 under **der-* “Spanne der Hand.” The grouping remained the standard handbook treatment for half a century.

The paper proceeds as follows. §2 sets out the formal foundation: Szemerényi’s Law, thematicization, and the Greek-internal model (εὐπάτωρ, δώτωρ). §3 develops the palm-as-measure cluster on **de-r-* (δάρις, *derna*, DERNV[, δῶρον · παλαιστή, and the δωρ-/δαρ- sibling derivation). §4 develops the arm-as-measure parallel on the s-stem **dóws-*. §5 gives the full etymology of δῶρον 2 with the derivation table. §6 addresses δῶρον 1 and the Vitruvian folk etymology. §§7–8 give the coexistence reading and address the departures. §§9–10 cover scope, falsifiability, and probability.

2. The Formal Foundation

The derivation of Greek δῶρον 2 and its open-palm cluster siblings uses a formal apparatus that is entirely Greek-internal and draws on sound laws and morphological processes uncontroversial in the standard handbooks. Three steps operate in sequence: Szemerényi’s Law on the consonant-stem r-stem nominative singular, yielding a lengthened nominative; thematicization of the resulting athematic stem; and the Greek-internal model provided by εὐπάτωρ and δώτωρ.

2.1 Szemerényi's Law and the εὐπάτωρ/δώτωρ Pattern

Greek preserves a productive class of consonant-stem r-stem nouns and adjectives in -ωρ whose long vowel in the nominative singular descends from Szemerényi's Law operating on pre-PIE **-Vrs#* sequences. The standard exemplars are uncontroversial. The adjective εὐπάτωρ “of a noble father” (accusative εὐπάτορα) descends from **eu-patōr-s*, with the nominative ending **-s* triggering compensatory lengthening and loss of the sibilant: pre-PIE **-Vrs#* → PIE **-Vr#*. The Homeric agent-noun δώτωρ “giver” (*Od.* 8.335) descends from **deh₃-tōr-s* by the same mechanism. The pattern extends throughout the Greek t-stem and r-stem agent-noun systems: ρήτωρ “speaker,” σωτήρ “savior,” πατήρ “father,” μήτηρ “mother,” θυγάτηρ “daughter.”

Szemerényi's Law itself is among the best-established sound rules of historical IE phonology. Lundquist and Yates (2018: 2083) summarize it as “PIE word-final **-VRs#* → **-VR#*” — a regular phonological rule on athematic consonant-stem nominatives whose stem-final segment is a sonorant followed by **-s*. Standard examples: **ph₂tīr* “father” from **ph₂tér-s*, **k(u)wōn* “dog” from **k(u)wén-s* (Szemerényi 1996: 113–119). Applied to the consonant-stem r-stem **de-r-* with o-grade nominative **do-r-s*, Szemerényi's Law produces **dōr* by exactly the mechanism that produces εὐπάτωρ from **eu-patōr-s*; the e-grade **de-r-s* yields **dēr* in parallel.

2.2 Thematicization of Athematic R-Stems

The lengthened consonant-stem nominative **dōr* produced by Szemerényi's Law on **do-r-s* is a bare athematic root-noun. Greek inherits a productive process by which such athematic stems are reanalyzed as bases for thematic derivation, yielding neuter o-stem nouns in **-o-m/*-o-n* alongside or in place of the inherited athematic form. This process is morphologically routine, documented in every reference grammar of historical IE morphology (Beekes 2011: 211–214; Fortson 2010: 132–134; Meier-Brügger 2003: 219–225).

Greek-internal parallels are abundant. δόρυ “spear, beam, tree-trunk” (genitive δόρατος) is a heteroclitic neuter that survives in attested Greek both as an athematic u-stem and as a thematicized formation in compound and derivative forms. γόνυ “knee” (genitive γόνατος) shows the same pattern. For δῶρον specifically, the Mycenaean form *do-ra* (KN Sd 4401, in the chariot tablets) attests δῶρα as a thematicized neuter plural already in the 13th century BCE, confirming that the surface morphology was in place by the Mycenaean date.

2.3 The Morphological Class

The framework's apparatus stands on the standard handbook treatment of Szemerényi's Law (Lundquist and Yates 2018: 2083; Szemerényi 1996: 113–119) and on the productive thematicization process documented across the IE reference grammars (Beekes 2011: 211–214; Fortson 2010: 132–134; Meier-Brügger 2003).

Beekes’s *Etymological Dictionary of Greek* (2010: 363) marks δῶρον 2 <?> (origin uncertain) and surveys rejected comparanda without a positive reconstruction; Beekes is a witness for the two-entry structure and the unresolved status of the etymology, not for the framework’s **de-r-* derivation.

3. The Palm-as-Measure Cluster on **de-r-*

This section develops the cognate forms lexicalized as the open palm or palm measurement: Hesychian δάρις/δάρειρ, Old Irish *derna*, Continental Celtic DERNV[, and the body-part sense of δῶρον 2. All four belong to the palm-as-measure subset of the framework’s body-part-as-measure cluster: the metrological extension of the r-stem arm-noun **de-r-* toward anatomical dimension, with the open palm (the distal portion of the arm-noun on the ARM → HAND grammaticalization of *The Hand That Gives* §4.3) lexicalized as the measurement unit. The cluster does not pass through giving or working; its driver is physical dimension, and its natural output is a measurement unit.

3.1 Hesychian Greek δάρις / δάρειρ (σπιθαμή)

The Hesychian lexicon glosses δάρις · σπιθαμή — “span,” the distance from the tip of the extended thumb to the tip of the extended little finger when the hand is spread wide. The standard Greek measurement σπιθαμή equals 12 fingers or half a cubit (Herodotus 2.106). Chantraine (1968: 305 s.v. δῶρον) gives the dialectal distribution: δάρις is Arcadian, and δάρειρ is its Laconian variant with the regular Laconian rhotacism. The open-palm cluster thus surfaces in three distinct Greek dialectal traditions: Arcadian δάρις, Laconian δάρειρ, and the wider Attic-Ionic and Doric δῶρον 2.

δάρις is an i-stem derivative on the bare e-grade r-stem **de-r-*, morphologically transparent: the bare arm-stem extended with the i-stem suffix, lexicalized as the measurement unit of the open hand. Frisk (*GEW* I 430–431) treats δάρις/δάρειρ on the same morphological footing. Chantraine groups δάρις, δάρειρ, and δῶρον 2 with Albanian *dorë* under a common etymon, citing Pokorny IEW 203. The Hesychian gloss is fringe lexicography — preserved in the marginal apparatus because the literary tradition adopted χεῖρ as the standard “hand” word — but measurement terminology is conservative, and the body-part anchor survived in it. Marstrander cited this gloss in his 1919 footnote.

3.2 Old Irish *derna* (Gaelic *deàrna*) “Palm”

Old Irish *derna* (Gaelic *deàrna*) “palm of the hand”, with alternant nominative *dernu* and genitive *dernand*. Pedersen (1913, vol. II: 110) places *dernu/dernand* in the same morphological paradigm as *gualu/gualand* “shoulder,” *Ánu/Ánann* (the goddess), and *áru/árann* “kidney” — a Celtic-internal classification alongside other alternating-stem nouns, made six years before Marstrander published, treating the form as a Celtic morphological observation rather than a comparative etymological

claim. The form is built on the bare e-grade r-stem **de-r-* with a feminine singulative **-nā-* extension, lexicalized as the open palm specifically. The Celtic palm-meaning is one of the daughter-language ARM → HAND specializations of *The Hand That Gives* §4.3.

3.3 Continental Celtic DERNV[at the Bronze of Novallas

The Celtiberian hospitium bronze published by Stifter (2022) preserves the form DERNV[in a Continental Celtic epigraphic context, related to the Insular Celtic *derna* “palm” cluster — the open-palm anchor in Continental Celtic epigraphic attestation. Stifter explicitly notes that the relationship between Old Irish *dernu* “palm” and *dorn* “fist” cannot be a simple case of ablaut, consistent with the framework’s reassignment of **durno-* to PIE **deru-/*drū-* (Paper 1 §7): the relationship between *derna* and **durno-* is not ablauting at all, but reflects two different PIE roots that converge phonologically in some Celtic outputs.

The same Bronze of Novallas preserves the Celtiberian directional preposition DV, the Celtiberian reflex of the Pre-PIE directional particle **de/do* developed in *The Hand That Gives* §3.2. Two derivational outputs of the framework’s bare **de-* — the body-part noun on the r-stem and the directional particle on the bare stem — surface in a single inscription.

3.4 δῶρον · παλαιστή: The Body-Part Sense (δῶρον 2)

The Hesychian gloss δῶρον · παλαιστή places “palm-breadth” alongside the gift-noun under the same lemma. The body-part and metrological senses are attested across Greek literature from the 8th century BCE forward: ἐκκαίδεκάδωρος “sixteen palm-widths long” in *Iliad* Δ 109; δεκάδωρος “ten palm-widths” in Hesiod *Works and Days* 426; the anatomical record of Pollux 2.144; and the architectural inscriptions from Miletus. LSJ groups all of these under δῶρον II (palm-related), distinct from δῶρον I (gift). Vitruvius’s folk-etymological story that motivated the ancient connection of palm and gift is addressed in §6. The full derivation of δῶρον 2 from **do-r-s* via Szemerényi and thematicization is developed in §5.

3.5 The δωρ-/δαρ- Relation: Sibling Derivations on **de-r-*

Schrijver (1995: 58) called the δωρ-/δαρ- relation “obscure”: the long *ō* of δῶρον 2 and the short *a* of δάρις cannot be derived from each other by any straightforward sound rule. Schrijver’s label is correct under the assumption that δωρ- and δαρ- stand in direct phonological relationship. The framework rejects this assumption.

On the framework’s analysis, δῶρον 2 and δάρις are sibling derivations on a shared base **de-r-*. δῶρον 2 is the thematicized output of the Szemerényi-derived o-grade nominative **dōr* (← **do-r-s*); δάρις is the i-stem extension of the bare e-grade stem **der-*. Different morphological operations on different ablaut grades of

the same r-stem. They are not derivable from each other because they are not phonological variants of the same word.

Three distinct morphological operations on the same r-stem base yield three distinct daughter-language outputs: δῶρον 2 (Greek, o-grade, Szemerényi-lengthened, thematicized); δάρις (Greek, e-grade, i-stem); *derna* (Celtic, e-grade, feminine singulative). Albanian *dorë* (Chavis and Claude 2026c) is a fourth: e-grade, Szemerényi-lengthened, *-eh₂- feminine extension. Schrijver's "obscure" verdict was the right verdict on the wrong question. The question is not "what sound rule links δωρ- and δαρ-" but "what derivational scaffold relates δῶρον 2, δάρις, *derna*, and *dorë*." The framework supplies that answer at the morphological level, where the relation is transparent as a sibling-derivation family on different ablaut grades.

4. The Arm-as-Measure Parallel: The s-Stem **dóws*- and the Cubit Family

The palm-as-measure cluster of §3 sits on the r-stem member **de-r*- of the body-part heteroclite class. The same class includes a sister s-stem formation **dóws*- "arm, fore-arm" (Pokorny IEW 226), which generates a parallel arm-as-measure cluster. The body-part heteroclite class therefore generates measurement vocabulary in both its members: the r-stem yields palm/span-class measures (~3–9 inches: the σπιθαμή of δάρις, the παλαιστή of δῶρον 2), the s-stem yields cubit-class measures (~18 inches). The metrology is a productive feature of the class, not an isolated feature of either heteroclite.

4.1 Sanskrit *dos* = 18 Inches (*Mātaṅgalīlā* 1.14)

Sanskrit *dos* is attested as a unit of 18 inches in the *Mātaṅgalīlā* 1.14, Nilakaṇṭha Sastri's elephant-treatise tradition. The 18-inch reading corresponds to a Sanskrit cubit-class measurement and aligns with the cross-cultural pattern of forearm-as-measurement-unit. The form *dos* is the bare s-stem arm-noun (sandhi-form *doh*, compound stem *dor*-) of Sanskrit *doṣ-/doṣán*- "arm, fore-arm" developed in *The Hand That Gives* §4.4. The cubit-class measure is the natural metrological output of an arm-noun: the cubit is the length from the elbow to the tip of the middle finger, approximately 18 inches in the standard ancient systems. The framework's reading is that the arm-noun **dóws*- generates the cubit-class measurement vocabulary by the natural lexical-metrological extension of body-part terms toward anatomical dimension, parallel to the r-stem arm-noun **de-r*- generating palm/span-class measurement vocabulary in §3.

4.2 *dor-jyā* "Arm-String" = Sine (*Sūryasiddhānta* 3.10)

Vedic *dor-jyā* "arm-string" is the Sanskrit mathematical term for the sine in the *Sūryasiddhānta* 3.10. The compound is geometrically explicit: the arm is the radius, the chord is the arm-string. The morphology is transparent: *dor-jyā* uses the

visarga-sandhi compound stem *dor-* of the s-stem arm-noun *doṣ-*, exactly the form developed in *The Hand That Gives* §4.4 for the productive Vedic compound system. The arm-noun's metrological function is carried forward into mathematical vocabulary on the same compound stem that produces the anatomical compounds (*dor-bāhū*, *dor-mūla*, *dor-daṇḍa*, *dor-nikartana*). The *dor-jyā* attestation places **dóws-* not only in linear-distance metrology but in geometric metrology: the arm is the radius of the trigonometric circle, the chord is the arm-string. The body-part heteroclite class generates measurement vocabulary not only for linear distance but for trigonometric ratio.

4.3 The Cross-IE Cubit Family

The cubit (and ell, half-cubit, hand-breadth) is a near-universal Indo-European metrological pattern, with the body-part lexicalized as the measurement unit across multiple branches and language families:

- Latin *cubitus* “elbow; cubit” — the cubit being the natural measure from elbow to fingertip
- Greek πῆχυς “forearm; cubit” — the same body-part-as-measure construction
- Hebrew *ammah*, literally “mother of the arm” — the cubit-class measurement explicitly anchored on the arm
- Sumerian *kuš* “cubit” — the same metrological category in the ancient Near East
- Old English *ell* “cubit, ell”; Old High German *elina* (whence English *elbow* < OE *eln-boga* “ell-bow”) — the metrological term feeding back into the body-part lexicon

The Sanskrit *dos* = 18 inches sits inside this cross-IE pattern. The smaller palm/span-class measurements — the σπιθαμή of δάρις, the παλαιστή of δῶρον 2, Old Irish *derna* “palm,” Continental Celtic DERNV[— sit at the other end of the same body-part metrology. The hand and arm together cover the relevant linear-distance scales: the cubit (~18 inches), the span (~9 inches, half-cubit), the palm (~3 inches, palm-breadth), the digit (~3/4 inch, finger-width). The body-part heteroclite class on **de-r-* / **dóws-* generates measurement vocabulary at multiple scales of this metrological range.

4.4 The Body-Part Heteroclite Class as the Productive PIE Pattern

The convergence of §§3–4 establishes that the body-part heteroclite class is a productive PIE pattern generating measurement vocabulary in both members. The r-stem **de-r-* produces palm/span-class measures across multiple branches; the s-stem **dóws-* produces cubit-class measures preserved most fully in Sanskrit. The two heteroclites are sister formations within the body-part heteroclite class; the metrology they generate is a productive feature of the class.

The pattern is also typologically natural. Body-part terms acquire measurement readings when the body part is habitually used as a measuring instrument (Hilpert 2007; Pattillo 2014). The framework’s contribution is the recognition that Greek δῶρον 2 belongs to this productive PIE pattern, not as an isolated lexeme of unknown origin.

5. The Etymology of Greek δῶρον 2

This section develops the framework’s central etymological claim: that Greek δῶρον 2 “palm-of-the-hand, palm-as-measure” descends from PIE **de-r-* via the Szemerényi-plus-thematicization apparatus of §2.

5.1 The Derivation

The framework derives δῶρον 2 from the consonant-stem r-stem **de-r-* (the body-part heteroclite arm-noun of *The Hand That Gives*, §4.3) via Szemerényi’s Law (§2.1) followed by thematicization (§2.2):

Step	Form	Mechanism	Status
1	<i>*do-r-s</i>	o-grade r-stem with nom. sg. <i>*-s</i>	Standard PIE inflection
2	<i>*dōr</i>	Szemerényi’s Law: pre-PIE <i>*-Vrs#</i> → PIE <i>*-Vr#</i>	Lundquist & Yates 2018: 2083; Szemerényi 1996: 113-119
3	<i>*dōr-o-n</i>	Thematicization of athematic r-stem; <i>*-os/*-on</i> extension	Standard PIE morphological process (Beekes 2011: 211-214; Fortson 2010: 132-134)
4	δῶρον	Greek surface phonology	Regular Greek; attested from Mycenaean <i>do-ra</i> (KN Sd 4401)

Table 1. The derivation of Greek δῶρον 2 from PIE **de-r-*. Each step uses only sound rules and morphological processes uncontroversial in the standard handbook treatments of PIE.

The derivation uses only sound rules and morphological processes that are uncontroversial in standard handbook treatments of PIE. Szemerényi’s Law on **-Vrs#* is in every IE textbook, and its operation on the consonant-stem r-stem **do-r-s* requires no further hypothesis. Thematicization of athematic r-stems with the **-os/*-om/*-on* formation is documented in the standard reference grammars; its application to the Szemerényi-derived nominative **dōr* is routine compositional morphology.

5.2 The Greek-Internal Model

The morphological pattern is the same one that produces εὐπάτωρ (from **eu-patōr-s*) and Homeric δώτωρ “giver” (from **deh₃-tōr-s*). In all three cases,

Szemerényi’s Law on a consonant-stem r-stem nominative produces the lengthened nominative. Where δῶτωρ and εὐπάτωρ surface as athematic Greek nouns in -ωρ, the framework’s **dōr* undergoes the further step of thematicization to surface as the neuter o-stem δῶρον. The thematicization step is what distinguishes δῶρον 2 (a neuter o-stem) from εὐπάτωρ and δῶτωρ (athematic r-stem adjectives and agent-nouns); the Szemerényi-derived nominative is the same morphological output in all three cases.

5.3 The Sibling Parallel with Albanian *dorë*

The framework’s **do-r-s* → **dōr* → **dōr-o-n* for Greek δῶρον 2 and the standalone companion paper’s **de-r-s* → **dēr* → **dēr-eh₂-* → **dēr-ā* → *dorë* for Albanian are structurally parallel derivations. Both start from a consonant-stem r-stem nominative; both apply Szemerényi’s Law to lengthen the nominative; both then extend the lengthened nominative with a derivational morpheme (thematic **-o-n* for Greek, feminine **-eh₂-* for Albanian). The only differences are the ablaut grade (o-grade for Greek δῶρον 2, e-grade for Albanian *dorë*) and the choice of extension morpheme. The convergence between the two outputs is morphological, not phonological: they share the same derivational scaffold applied at PIE date in different grades.

5.4 Mycenaean Evidence

The Linear B form *do-ra* (KN Sd 4401, in the chariot tablets) attests δῶρα as a thematicized neuter plural already in the 13th century BCE. The chariot inventory context does not disambiguate between gift-δῶρα and palm-δῶρα, but it confirms that the thematicized neuter morphology was in place by the Mycenaean date, consistent with the framework’s PIE-date thematicization.

5.5 The Greek Lexical Record for δῶρον 2

The body-part and metrological senses are attested from the earliest Greek poetry forward:

- *Iliad* Δ 109: ἑκκαίδεκάδωρος “sixteen palm-widths long” — of Pandaros’s bow horns
- Hesiod *Works and Days* 426: δεκάδωρος “ten palm-widths” — of an axle
- Hesychian gloss δῶρον · παλαιστή — the body-part anchor in lexicographic tradition
- Pollux *Onomasticon* 2.144: δῶρον as the front of the palm in a strictly anatomical context
- Miletus inscriptions: δῶρον as a standard architectural unit

LSJ groups all of these under δῶρον II (palm-related), distinct from δῶρον I (gift). The framework’s etymology applies to all of these uses together, since they are all uses of the same lexeme δῶρον 2.

5.6 What the Framework Supplies that Frisk and Beekes Left Open

Frisk marks δῶρον 2 as “Origin: XX [etym. unknown]” while still citing La Piana for the Albanian comparison. Beekes (2010: 363) marks it <?> and his ETYM section is a survey of rejected comparanda without a positive reconstruction. The framework supplies the full derivation both left open: Szemerényi’s Law on **do-r-s* → **dōr*, thematicization to **dōr-o-n*, surface as δῶρον. The PIE root underlying δῶρον 2 is **de-r-* (the body-part heteroclite arm-noun), not **deh₃-* (the give-verb). Frisk’s “Origin: XX” and Beekes’s “<?>” both reflect an etymology that is available through Szemerényi and thematicization once the right root is identified.

6. δῶρον 1 “Gift” and the Vitruvian Folk Etymology

The framework adopts the standard derivation of Greek δῶρον 1 “gift” from PIE **deh₃-ro-* (the give-verb **deh₃-* extended by the nominalizing suffix **-ro-*) without modification. The reconstruction is uncontested and supported by an unbroken cross-branch cognate set: Greek δῶρον “gift,” Latin *dōnum* (from **deh₃-no-*), Sanskrit *dāna-*, Armenian *tur*, OCS *darъ*, Old Irish *dán*, Welsh *dawn*. The derivational scaffolding is in every standard reference work (Frisk *GEW* I 433; Beekes 2010: 362; Pokorny *IEW* 223–226). For Greek δῶρον 1 specifically, the standard analysis stands.

Under the framework’s analysis, Vitruvius (*De architectura* 2.3.3), the Hesychian gloss δῶρον · παλαιστή, and Pollux *Onomasticon* 2.144 all attest the synchronic homophony of two etymologically distinct words in attested Greek. Vitruvius’s etymological story — that the palm is called “gift” because giving is what the open hand does — is folk etymology of the standard kind: speakers confronted with two homophones in their language construct a semantic bridge to motivate the perceived connection. The construction Vitruvius records is not derivational evidence; it is evidence that the homophony existed and was noticed.

The English parallel is exact: *ear* “organ of hearing” (from PIE **h₂ous-*) and *ear* “spike of grain” (from PIE **h₂ék-* “sharp”) are synchronically homophonous words with entirely different etymological histories; folk-etymological stories link them. The etymological histories are independently established by the comparative method, and the folk etymology is recognized as a synchronic phenomenon that does not bear on the historical reconstruction. The same analytical posture applies to the two δῶρα. The framework does not deny the cross-linguistic HAND → GIVE grammaticalization pathway (documented by Heine and Kuteva 2002) — that pathway motivates the Pre-PIE **de-* unification at the deeper level of *The Hand That Gives* §2 — but it does not run through δῶρον 2 → δῶρον 1 in the surface Greek lexicon.

7. The Coexistence Reading

The framework adopts the coexistence reading with respect to the other PIE hand-root reconstructions. PIE **ǵʰesr-/or-/to-* (Pokorny IEW 447) is accepted in full as the source of the cross-IE “hand-as-grasp” word: Greek χεῖρ, Hittite *kessar*, Tocharian A *tsar* / B *šar*, Armenian *jeṛn*; via **ǵʰesto-*, Sanskrit *hásta-*, Avestan *zasta-*, Old Persian *dasta-*. PIE **deh₃-* is accepted in full as the source of the cross-IE give-verb and the gift-noun cluster on **deh₃-ro-/no-*. The framework claims only that **de-r-* is a separate PIE etymon on a different morphological base, centered on the arm and its derivative palm-meaning, with its own coherent derivational outputs. The three etyma likely had specialized senses in early PIE: **ǵʰesr-* the hand as joint/grasp; **de-r-* the arm and its derivative open palm; **deh₃-* the verbal act of extension-toward-recipient.

Pokorny kept **der-* (IEW 203), **dóws-* (IEW 226), **ǵʰesor-* (IEW 447), and the give-verb (under **dō-* at IEW 223–226) as separate entries; this paper is methodologically aligned with that pre-existing taxonomy. Specialists who reject the **de-r-* / **ǵʰesr-* separation can evaluate the framework’s claims about the δῶρον 2 cluster independently; the framework’s contribution is the recovery of **de-r-* as a distinct PIE etymon with its own derivational outputs, not a displacement of the χεῖρ family.

8. The Schrijver 1995 “Obscure” Verdict and the Beekes 2010 Celtic **u* Exclusion

Two late-twentieth-century departures targeted the Pokorny IEW 203 grouping. Schrijver (1995) declared the δῶρ-/δάρ- relation “obscure”; Beekes (2010), following Matasóvič (2009), excluded the Celtic forms on Celtic **u* grounds.

Schrijver 1995. The “obscure” verdict resolves under the sibling-derivation analysis of §3.5. δῶρον 2 and δάρις are not phonological variants of one form — they are sibling derivations on the shared r-stem base **de-r-*. The framework’s contribution is the relocation of the relation from the phonological level (where Schrijver was right that no straightforward sound rule links the forms) to the morphological level (where the relation is transparent as a sibling-derivation pair on different ablaut grades).

Beekes 2010 / Matasóvič 2009 on Celtic **u*. These objections target the Celtic fist-cluster (**durno-*, *dorn/dwrn*, Breton *dorn*, Latvian *dūre*), not the open-palm cluster. The framework separately concludes that the fist-cluster derives from PIE **deru-/drū-* “firm, hard, oak, tree” rather than from **de-r-* “arm, open palm,” with the Celtic **u* as the regular zero-grade outcome of the tree-root (Paper 1 §7). A fist is the hand in its hardened, firm posture; the etymological homeland of the Celtic fist-word is the firm-hard cluster. This reassignment removes the Celtic **u* problem from the present paper entirely. What remains is the open-palm cluster proper (δῶρον 2, δάρις, *derna*, DERNV[, *dorē*), all on the e-grade or o-grade r-stem **de-r-* without any **u*-vocalism. The Beekes 2010 exclusion of Celtic **u* and the

Matasóvič 2009 substrate hypothesis are therefore not applicable to the framework's central claims in this paper.

9. Scope and Falsifiability

Scope

The framework does not introduce a new PIE root. The body-part heteroclite arm-noun **de-r-* is developed in the companion paper *The Hand That Gives*; the present paper develops the body-part-as-measure cluster issuing from it. The framework adopts the coexistence reading (§7) with respect to **ĝh^hesr-* and the standard **deh₃-*. The standard derivation of Greek δῶρον 1 from **deh₃-ro-* is preserved without modification (§6). The Celtic fist-cluster (**durno-*, *dorn/dwrn*, Latvian *dūre*) is reassigned to PIE **deru-/*drū-* (Paper 1 §7) and is not part of the present paper's open-palm or body-part-as-measure cluster.

Falsifiability

The framework is falsifiable in four distinct ways. First, a positive demonstration that δῶρον 2 “palm” descends from **deh₃-* by some derivational path (heteroclitic apparatus, semantic extension from gift to palm, or alternative morphology) would remove the framework's central contribution at §5. Second, demonstrated within-branch variation in the δῶρ-/δάρ- distribution not captured by the sibling-derivation analysis (§3.5) would reopen Schrijver's “obscure” verdict. Third, a positive Anatolian etymology for *tarawar/tarna-* derived from a non-**de-r-/non-*deh₃-* root with independent Anatolian-internal support would weaken the cross-branch coherence of the cluster, though it would not directly affect the §5 derivation of δῶρον 2. Fourth, demonstration that the Sanskrit *dos* = 18 inches and *dor-jyā* = sine attestations belong to a different etymological root from the s-stem arm-noun *doṣ-/doṣán-* would weaken the §4 arm-as-measure parallel, though it would not directly affect the §5 derivation.

10. Probability Assessment

We assess the probability of this paper's central thesis — that Greek δῶρον 2 “palm-of-the-hand” descends from the consonant-stem r-stem **de-r-* via Szemerényi plus thematicization, that this form belongs with δάρης, *derna*, DERNV[, and Albanian *dorë* in a palm-as-measure cluster on **de-r-*, and that the s-stem sister **dóws-* generates a parallel arm-as-measure cluster — at approximately 50–60%, conditional on Paper 1's framework being correct. The conditional probability is higher than Paper 1's own 45–55% because this paper's claim is narrower in scope: it does not depend on the Anatolian take-verb analysis or the Luwian **lā-* problem. A reader who accepts the body-part heteroclite arm-noun **de-r-* and its s-stem sister **dóws-* but reserves judgment on the take-verb material can still accept this paper's claims about the body-part-as-measure cluster.

The probability sits in the middle-to-upper range of the disputed-but-plausible interval because the framework's claim is narrowly scoped, uses only standard handbook morphology (Szemerényi plus thematicization), and supplies an etymology for a form that Frisk already marks as origin unknown. The framework does not displace the standard derivation of δῶρον 1 from **deh₃-ro-*; it does not require the Clayton/Blažek heteroclitic apparatus; it does not run through the contested Anatolian take-verb or the Luwian **lā-* problem. A specialist open to revisiting Frisk's "Origin: XX" entry on δῶρον 2 has no friction in evaluating the framework's central claim independently.

In Favor

- Frisk and Beekes already treat δῶρον 1 and δῶρον 2 as separate entries. The framework's two-entry treatment is not novel; what is novel is the etymology supplied for the second entry.
- The morphological apparatus — Szemerényi's Law on consonant-stem r-stem nominatives, followed by thematicization — uses only standard handbook tools (Lundquist and Yates 2018: 2083; Szemerényi 1996: 113–119; Beekes 2011: 211–214; Fortson 2010: 132–134).
- The open-palm cluster around δῶρον 2 (δάρις, *derma*, DERNV[]) is morphologically coherent: different extensions on the same r-stem base in different branches.
- The arm-as-measure parallel on **dóws-* (Sanskrit *dos* = 18 inches, *dor-jyā* = sine) places δῶρον 2 inside a productive PIE body-part-as-measure pattern.
- The cross-IE cubit family (Latin *cubitum*, Greek πῆχυς, Hebrew *ammah*, Old English *ell*) is a near-universal Indo-European metrological pattern; the Sanskrit *dos* sits inside it.
- Schrijver's "obscure" verdict resolves under the sibling-derivation analysis (§3.5), and the Beekes 2010 Celtic **u* exclusion is addressed by the reassignment of the fist-cluster to **deru-/drū-* (Paper 1 §7).
- Mycenaean *do-ra* (KN Sd 4401) confirms that the thematicized neuter morphology was in place by the 13th century BCE.

Against

- The two-δῶρα analysis remains a minority view; the post-Schrijver consensus has favored treating the gift-side and palm-side as derivationally unified, or treating the palm-side as a secondary semantic development of the gift-side.
- The body-part heteroclitic class as a productive PIE pattern with both r-stem and s-stem members is not standard in current handbook treatments; Pokorny kept **der-* (IEW 203) and **dóws-* (IEW 226) as separate entries without explicitly classifying them as sister heteroclitics.

- The chief residue is the irregular plural *duar* of Albanian *dorë* (treated in the standalone *dorë* paper), which no current reconstruction handles transparently. This residue does not affect the present paper’s derivation of δῶρον 2 but bears on the broader r-stem cluster.

Project-Level Marginal Probability

The marginal probability of the broader project — integrating conditional probabilities across the companion papers (Paper 1 *The Hand That Gives*, the standalone Albanian *dorë* paper, and the present paper) — sits in the 2–5% range. Much of the evidence supports multiple strata simultaneously rather than each stratum independently; the honest marginal range is toward the lower end of that interval. We regard the central claim as more likely wrong than right at the marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

11. Authorship and Contributions

This paper is co-authored by J.M. Chavis (human) and Claude (Anthropic AI; primarily Claude Opus 4.7).

Chavis’s contributions. The central hypothesis originates with Chavis: the identification of the dual attestation of Greek δῶρον — gift and palm-breadth — as evidence of two etymologically distinct words was the project’s founding insight. Chavis directed the research agenda at every stage: the architectural decision to split the previous combined Paper 1 into two papers (Paper 1 *The Hand That Gives* on the arm-extension and Anatolian material; the present companion on δῶρον 2 and the body-part-as-measure cluster); the editorial decision to include the arm-as-measure material from the s-stem **dóws-* as parallel to the palm-as-measure cluster on the r-stem; the redirection of the Albanian *dorë* defense to its own standalone paper; the reframing of the Celtic fist-cluster onto the **deru-/*drū-* foundation; and the methodological pivot to the coexistence reading. All fundamental decisions about scope, framing, and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude’s contributions are primarily in source synthesis, exposition, and analytical development. Claude carried out the literature reading, cross-branch corroboration searches, and technical drafting under Chavis’s direction. Claude engaged with Lundquist and Yates 2018, Szemerényi 1996, Beekes 2011, Fortson 2010, Meier-Brügger 2003, Hilpert 2007, Pattillo 2014, Stifter 2022, Pedersen 1913, Chantraine 1968, Frisk 1960–72, Mayrhofer EWAia, and the Sanskrit lexicographical and metrological sources (*Mātaṅgalīlā*, *Sūryasiddhānta*, Cologne lexicons, wisdomlib); examined Pokorny IEW pp. 203, 226, and 447 directly; verified the Mycenaean Linear B attestation *do-ra* (KN Sd 4401); and integrated the architectural alignment with the companion papers. The

probability estimates are Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management.

Both authors take responsibility for the framework and the specific claims made in this paper.

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